Finite certificates for Hamiltonian thresholds in planar recolouring graphs

Abstract

This supplement contains the literal local certificates used in *Hamiltonian and Hamilton-connected thresholds for planar recolouring graphs*. The accompanying verification programs reconstruct the finite local graphs and check the legality, coverage, adjacency, symmetry classes, and required interface and protected-edge conditions of the supplied certificates. The inductions and arbitrary-order restoration arguments are proved in the article.

A Adjacent-pair endpoint certificates

For the article's fibre $F(A, B)$, write ab for (a, b) and use type $(|A|, |B|, |A \cap B|)$. Each endpoint word visits every state once; successive states differ in one coordinate. Table A.1 includes list-preserving colour permutations, allowable coordinate exchange and path reversal. Exact subset dynamic programming checks all feasible and excluded endpoint pairs.

Table A.1: Adjacent-pair endpoint certificates and their symmetry coverage.

Type	$\|V(F)\|$	Words	Feasible unordered pairs
331	8	6	28
332	7	7	19
333	6	1	6
342	10	16	45
343	9	8	36
443	13	11	78
444	12	4	66

A.1 Edge-covering cycles

The columns give the type, lists separated by a slash, and one cyclic word. Its list-preserving symmetry orbit, including allowable coordinate exchange, covers every edge.

```
331   012/034    03 04 14 10 20 24 23 13
332   012/013    23 03 01 21 20 10 13
333   012/012    10 20 21 01 02 12
342   012/0134   24 04 01 03 13 14 10 20 21 23
343   012/0123   12 10 20 21 01 02 03 23 13
443   0123/0124  32 12 10 20 21 01 02 04 24 14 34 31 30
444   0123/0123  23 21 20 10 12 02 32 30 31 01 03 13
```

The 53 endpoint words. The list pairs are given in the preceding cycle table.

For 331:

```
03 13 23 24 20 10 14 04
03 23 20 24 04 14 13 10
03 23 20 24 04 14 10 13
03 23 13 10 20 24 04 14
13 10 20 24 23 03 04 14
13 14 10 20 23 03 04 24
```

For 332:

```
01 21 23 20 10 13 03
01 21 20 23 03 13 10
01 03 23 21 20 10 13
01 21 20 10 13 03 23
03 01 21 23 20 10 13
03 01 21 23 13 10 20
03 01 21 20 10 13 23
```

For 333:

```
01 21 20 10 12 02
```

For 342:

```
01 21 24 20 23 13 10 14 04 03
01 21 23 20 24 14 04 03 13 10
01 21 24 23 20 10 14 04 03 13
01 21 23 24 14 04 03 13 10 20
01 04 14 24 23 03 13 10 20 21
01 21 24 20 10 13 14 04 03 23
03 23 24 14 13 10 20 21 01 04
03 23 24 20 21 01 04 14 10 13
03 13 10 20 24 23 21 01 04 14
03 23 24 21 01 04 14 13 10 20
03 13 23 24 20 10 14 04 01 21
03 01 21 20 24 04 14 10 13 23
03 23 13 14 10 20 21 01 04 24
20 24 23 13 10 14 04 03 01 21
20 10 13 14 04 24 21 01 03 23
23 21 20 10 14 13 03 01 04 24
```

For 343:

```
01 21 23 20 10 12 13 03 02
01 21 20 23 13 10 12 02 03
01 21 20 23 13 03 02 12 10
01 21 23 20 10 13 03 02 12
01 21 23 20 10 12 02 03 13
01 02 03 23 13 12 10 20 21
01 21 20 10 13 12 02 03 23
03 23 20 21 01 02 12 10 13
```

For 443:

```
01 31 34 24 21 20 30 32 12 10 14 04 02
01 21 31 34 32 30 20 24 14 10 12 02 04
01 21 31 34 32 30 20 24 14 04 02 12 10
01 31 32 30 34 24 21 20 10 14 04 02 12
01 31 32 30 34 24 21 20 10 12 02 04 14
01 21 20 30 32 31 34 14 10 12 02 04 24
01 21 31 32 30 20 24 14 10 12 02 04 34
04 24 34 32 31 30 20 21 01 02 12 10 14
04 34 14 24 20 21 01 31 32 02 12 10 30
04 34 32 30 20 21 24 14 10 12 02 01 31
04 24 20 30 32 31 21 01 02 12 10 14 34
```

For 444:

```
01 31 21 23 20 30 32 12 10 13 03 02
01 21 31 32 30 20 23 13 03 02 12 10
01 21 31 32 30 10 20 23 13 03 02 12
01 21 31 32 30 20 10 13 12 02 03 23
```

B Certificates for the $(2, 3, 2)$-path restoration

The fibres B_t, allowed colour images and path reversals, and special edge sets P_t are defined in the main text. The following literal cycles supply the selected-layer identities used in the safe-common-edge lemma in the article.

B.1 The three normalized cycles

In the cyclic words below the coordinates are uvw. The rows for $001, 010, 012$ specify D_E, D_M, D_A, respectively.

001	120 123 323 313 213 210 230 130 132 232 212 312 310 320
010	101 121 321 323 123 103 303 302 301
	201 203 202 102 132 232 231 131
012	131 101 103 203 303 301 201 231
	230 130 120 123 323 320 321 121

(B.1)

B.2 Selected-layer identities and capacities

Order the end-, middle-, and all-distinct types as $E < M < A$. For distinct a, b, c, d, let $E(aab), M_0(aba), A(abc)$ be the images of D_E, D_M, D_A under $(0, 1, 2, 3) \mapsto (a, b, c, d)$, and let M_1 be the reversal of M_0. Unused-colour interchange fixes E, M_0, M_1; reversal followed by endpoint-colour interchange fixes A. Thus E, A each have one allowed edge set and M has the two stated sets. Up to these symmetries, the good transitions to a type no smaller are

$$E(aab) : aac, abb, acb, bab, cab; \quad M(aba) : aca, cba; \quad A(abc) : abd, adc.$$

These exhaust one-coordinate changes; bad targets and smaller types are excluded, and unused colours and the two ends are identified by the stated symmetries.

In $E(aab)$, the middle-b layer is $2P_3$ and the middle-c, d squares are P_4. In $M_0(aba)$, the middle-a layer is P_8 with isolated bab, and the middle-c, d squares are P_4; reverse for M_1. In $A(abc)$, the middle-a, c layers are P_6 and the middle-d square is P_3 with isolated bdb. These are subgraphs of the respective selected cycles.

Each cycle has four middle-changing edges. For E, middle colour b joins c, d twice each, with last coordinate a twice and c, d once each. For M_0, middle colour a joins c, d twice each and every edge has first coordinate b; reverse for M_1. For A, middle colour d joins a, c twice each, with first coordinate in $\{b, c\}$ and last coordinate in $\{a, b\}$. These paths and counts are read directly from (B.1).

For common endpoint lists L, R and distinct allowed middle colours v, w, the layer H_v is $K_{|L\setminus\{v\}|} \square K_{|R\setminus\{v\}|}$, and there are $|L\setminus\{v, w\}|\,|R\setminus\{v, w\}|$ edges between middle colours v, w. Removing the special edges gives the safe universe size m; the original cyclic words give the retained counts x, y. The following seven families supply the capacities used in the proof:

$t \to s$	blocks	(m, x, y)
$aab \to aac$	H_*	$(4, 3, 3)$
$aab \to abb$	H_c	$(2, 2, 2)$
$aab \to acb$	H_*	$(2, 2, 2)$
$aab \to cab$	H_c, H_d	$(3, 2, 2), (1, 1, 1)$
$aba \to aca$	H_d	$(4, 3, 3)$
$abc \to abd$	H_*	$(6, 4, 4)$
$abc \to adc$	H_a, H_c	$(9, 5, 5)$ each

C Literal pentagon cycles, path blocks and common-port seeds

Five digits record the five colours in cyclic vertex order. In six-digit words the first digit is the path layer. The two constant-type component cycles are printed separately. Cycle words are cyclic; the 51-state and 47-state paths for 00112 are linear. A port is a pair of five-digit words. The ordered path blocks use the specified cycles and the two additional identities below; no eleven-type factor-word list is needed. These literal identities are part of the proof, not solver verdicts.

Table C.1: Ordered pairs for 00001. Each pair $a_i b_i$ is a block; the connector to the next block is $a_i a_{i+1}$ when $\epsilon_i = 0$ and $b_i a_{i+1}$ when $\epsilon_i = 1$.

i	a_i	b_i	ϵ_i	i	a_i	b_i	ϵ_i
1	12132	32132	0	12	21310	31310	1
2	12130	32130	0	13	31312	32312	0
3	12120	32120	0	14	31212	31232	1
4	13120	23120	0	15	31230	21230	1
5	13123	23123	0	16	21210	31210	0
6	12123	12323	1	17	21213	23213	1
7	12313	12312	1	18	23210	13210	1
8	12310	32310	0	19	13212	13213	0
9	12320	32320	1	20	13232	13132	1
10	31320	21320	1	21	13130	23130	0
11	21323	21313	1	22	13230	23230	-

Path for 00112 (51 words).

12303 12323 12023 13023 13020 12020 12320 32320 31320 21320
21323 21303 21203 21201 21301 21321 31321 32321 32301 31301
31201 31231 31230 21230 21231 21031 21030 31030 31031 32031
32021 31021 31020 32020 32030 12030 13030 13230 23230 23030
23020 21020 21021 21023 23023 23021 23031 23231 23201 23203
13203

Constant-boundary cycles (30 words).

12123 12323 12313 12312 32312 31312 31212 31232 31231 21231 23231 23131 23121 23123 13123
12132 13132 13232 13212 13213 23213 21213 21313 21323 21321 31321 32321 32121 32131 32132

Cycle 00011 (42 words).

12102 13102 13103 23103 23123 13123 13120 23120 23130 13130
13230 23230 21230 31230 31232 31202 31302 32302 12302 12303
12103 12123 12323 12320 32320 31320 21320 21323 21303 21203
23203 13203 13202 13232 13132 12132 12130 12120 32120 32130
32132 32102

Cycle 00012 (47 words).

12103 12123 13123 13120 12120 32120 32130 12130 13130 13230
23230 21230 31230 31231 21231 23231 23201 23101 23131 23130
23120 23121 23123 23103 13103 13203 23203 21203 21201 31201
31301 32301 32101 32131 32121 32321 31321 21321 21301 21303
21323 21320 31320 32320 12320 12323 12303

Cycle 00101 (64 words).

12010 12020 32020 31020 21020 23020 13020 13010 23010 23210
13210 13212 13232 13032 13012 12012 32012 31012 31312 32312
12312 12310 32310 31310 21310 21320 31320 32320 12320 12323
12313 12013 12023 13023 23023 21023 21323 21313 21013 23013
13013 13213 23213 21213 21210 31210 31212 31232 31032 32032
12032 12030 32030 31030 31230 21230 23230 13230 13030 23030
21030 21010 31010 32010

Cycle 00102 (63 words).

12010 12030 32030 31030 31230 21230 23230 13230 13030 23030
21030 21031 23031 23231 21231 31231 31031 32031 32021 32321
31321 21321 21320 31320 32320 12320 12310 32310 31310 21310
21313 21323 21023 23023 13023 13020 13010 23010 21010 21020
23020 23021 21021 31021 31020 32020 12020 12023 12323 12313
12013 13013 23013 21013 21213 23213 13213 13210 23210 21210
31210 31010 32010

Cycle 00121 (58 words).

12010 13010 23010 23210 13210 13230 13030 12030 32030 31030
31032 32032 12032 13032 13232 13202 13212 13012 12012 32012
31012 31312 32312 12312 12310 32310 31310 21310 21210 21213
23213 13213 13203 23203 21203 21303 21313 21013 23013 13013
12013 12313 12303 12302 32302 31302 31202 31232 31212 31210
31230 21230 23230 23030 21030 21010 31010 32010

Cycle 00123 (57 words).

12010 12310 32310 31310 21310 21010 23010 13010 13210 13212
13202 13232 13032 13012 12012 32012 31012 31212 31232 31032
32032 12032 12030 32030 32031 31031 21031 21030 31030 31230
21230 21231 31231 31201 31202 31302 31312 32312 12312 12302
32302 32301 31301 21301 21201 23201 23231 23031 23030 13030
13230 23230 23210 21210 31210 31010 32010

Cycle 01012 (68 words).

10103 10203 13203 23203 20203 20303 10303 12303 12103 12123
10123 13123 23123 20123 20323 10323 12323 12320 32320 30320
10320 20320 20120 23120 13120 10120 30120 32120 12120 12130
32130 30130 10130 10230 30230 20230 23230 13230 13130 23130
20130 20131 23131 23231 20231 30231 30131 32131 32101 32301
30301 20301 20321 30321 32321 32121 30121 20121 23121 23101
23201 20201 30201 30101 20101 20103 23103 13103

Cycle 01023 (66 words).

10102 10302 30302 32302 12302 12312 10312 30312 32312 32310
12310 10310 30310 20310 20210 30210 10210 10212 30212 30202
10202 13202 13102 13132 13232 13212 13210 23210 23230 23130
23131 23231 23201 23101 20101 30101 30201 20201 20301 30301
32301 32101 32131 30131 30231 20231 20131 20130 20230 30230
10230 13230 13130 12130 32130 30130 10130 10132 10232 30232
30132 32132 12132 12102 32102 30102

C2 for 00112 (102 words).

```
012020 012030 032030 032031 032021 031021 021021 021031 031031
031231 031201 031301 032301 032321 032320 012320 012323 012303
112303 112323 112320 132320 132321 132301 131301 131201 131231
131230 031230 031030 021030 021230 121230 121030 131030 131020
121020 123020 123023 123021 123031 123030 123230 113230 113030
113020 113023 112023 012023 013023 013020 013030 013230 023230
023030 023031 023231 123231 121231 021231 021201 021301 021321
031321 131321 131021 131031 121031 121021 121023 121323 121303
021303 021203 121203 123203 113203 013203 023203 023201 123201
121201 121301 121321 121320 131320 031320 021320 021323 021023
023023 023021 023020 021020 031020 032020 132020 132021 132031
132030 112030 112020
```

Three-row insertion rule. Let C_2 be a Hamilton cycle of $A\square P_2$, with row digits $0, 1$. Suppose its second row contains three disjoint edges u_iv_i, and the states of A partition into three paths with respective endpoints u_i, v_i. Replace each cut edge $(1, u_i)(1, v_i)$ by its path in row 2 and the cross edges at both endpoints. These paths are internally disjoint and cover the new row, so they subdivide three edges of C_2 into one Hamilton cycle of $A\square P_3$. If D_A occurs in both old rows and in the new paths, and avoids the three cut edges, it survives in every row of the derived seed.

Common ports 00112 (13 edges).

```
12023-13023 12320-32320 13020-13023 13030-13230
13230-23230 21020-23020 21020-31020 21030-21230
21030-31030 21301-21321 31201-31301 32030-32031
32301-32321
```

Generating the three-row seed for 00112. The following path P contains forty-seven distinct states and all thirteen ports above. Together with $12303, 12323, 13203, 23203$, its vertices are precisely the legal states of 00112.

```
21201 21203 21303 21301 21321 21320 21323 21023 23023 23021
23031 23030 23230 13230 13030 12030 32030 32031 32021 32020
31020 21020 23020 13020 13023 12023 12020 12320 32320 31320
31321 32321 32301 31301 31201 31231 31230 21230 21030 31030
31031 31021 21021 21031 21231 23231 23201
```

In row 1 of the printed C_2, cut

$$12303\text{–}12323, \qquad 13203\text{–}23203, \qquad 21201\text{–}23201.$$

The new row uses the first two edges as its two-point paths and P as the third path. All cut edges avoid D_A, which occurs in both rows of C_2 and in P. The insertion rule therefore gives the required three-row seed with all thirteen common ports.

D Literal protected-pentagon certificates

Boundary digit 4 denotes the full list. Colour words use only 0,1,2,3. In six-digit cycle words the first digit is the path layer. The first two pool edges for a Hamilton component join two or three path rows; any two remaining pool edges supply block-chain reserves. The protected set is the specified cycle edge set, or COMMON set for a hard type, minus POOL. The earlier appendix supplies the ordinary cycles, the two constant-type 15-cycles, and the 00112 seeds. An interface record gives source type:component, position, new colour, normalized target type:component, and the actual common edge. Positions start at zero. Two records per line are separated by semicolons.

PC2 for 00001 (88 words).

```
012120 012130 112130 112120 132120 132130 132132 112132 113132
113232 113230 113210 123210 123213 113213 113212 013212 013213
023213 021213 021210 023210 013210 013230 013232 013132 012132
032132 032130 032120 032320 032310 012310 012320 012323 012313
012312 032312 031312 031310 031210 031212 031232 031230 021230
023230 123230 121230 131230 131232 131212 131210 121210 121213
121313 121323 021323 021313 021310 121310 121320 021320 031320
131320 132320 112320 112310 132310 131310 131312 132312 112312
112313 112323 112123 012123 013123 023123 123123 113123 113120
123120 123130 113130 013130 023130 023120 013120
```

Common ports 00001 (13 edges).

```
12132-13132 12132-32132 12310-32310 13120-23120
13123-23123 13210-23210 13230-13232 21210-21213
21230-23230 21230-31230 21313-21323 31212-31232
32130-32132
```

Generating the three-row seed for 00001. The following path P contains forty distinct states and all thirteen ports above. Together with $12312, 12313, 13212, 13213$, its vertices are precisely the legal states of 00001.

```
13210 23210 23213 21213 21210 21310 21313 21323 21320 31320
32320 32120 12120 13120 23120 23123 13123 12123 12323 12320
12310 32310 32312 31312 31310 31210 31212 31232 31230 21230
23230 23130 13130 12130 32130 32132 12132 13132 13232 13230
```

In row 1 of the printed C_2, cut

$$12312\text{–}12313, \qquad 13212\text{–}13213, \qquad 13210\text{–}13230.$$

The new row uses the first two edges and P. All cuts avoid D_A, which occurs in both rows of C_2 and in P. Apply the insertion rule in Appendix C to obtain the three-row seed, retaining every listed common port.

Cycle for 00004 (54 words).

```
12120 12130 12132 13132 13130 23130 23230 23210 13210 13230
13232 13212 13213 23213 21213 21210 31210 31212 31232 31231
31230 21230 21231 23231 23131 23121 23123 23120 13120 13123
12123 12323 12313 12312 12310 32310 32312 31312 31310 21310
21313 21323 21321 21320 31320 31321 32321 32121 32131 32132
32130 32120 32320 12320
```

Cycle for 00014 (60 words).

12102 13102 13132 13232 13202 13203 13103 12103 12123 13123
13120 12120 32120 32320 12320 12323 12303 12302 32302 31302
31202 31232 31230 21230 23230 13230 13130 23130 23120 23121
23123 23103 23203 23201 23101 23131 23231 21231 31231 31201
21201 21203 21303 21323 21320 31320 31321 21321 21301 31301
32301 32321 32121 32101 32131 32130 12130 12132 32132 32102

Cycle for 00104 (78 words).

12010 12030 32030 31030 31230 21230 23230 13230 13030 23030
21030 21031 23031 23231 21231 31231 31031 32031 32021 32321
32320 12320 12323 12313 12013 12023 13023 23023 21023 21020
31020 32020 12020 13020 23020 23021 21021 31021 31321 31320
21320 21321 21323 21313 21310 31310 32310 12310 12312 32312
31312 31012 32012 12012 13012 13013 23013 21013 21213 23213
13213 13212 13232 13032 12032 32032 31032 31232 31212 31210
21210 23210 13210 13010 23010 21010 31010 32010

Cycle for 00114 (62 words).

12020 12320 32320 31320 21320 21323 21023 23023 23021 23031
21031 21021 21020 21030 21230 31230 31030 32030 12030 13030
13230 23230 23030 23020 13020 13023 12023 12323 12303 12302
32302 31302 31202 31232 31032 32032 12032 13032 13232 13202
13203 23203 21203 21303 21301 21321 31321 32321 32301 31301
31201 21201 23201 23231 21231 31231 31031 32031 32021 31021
31020 32020

Cycle for 00124 (71 words).

12010 12030 32030 32031 31031 31231 21231 21031 21030 31030
31230 21230 23230 13230 13030 23030 23031 23231 23201 23203
21203 21303 21301 21201 31201 31202 31302 31301 32301 32302
12302 12303 12313 12013 13013 13010 13210 13212 13012 12012
32012 31012 31312 32312 12312 12310 32310 31310 31210 31212
31232 31032 32032 12032 13032 13232 13202 13203 13213 23213
21213 21313 21013 23013 23010 23210 21210 21310 21010 31010
32010

Cycle for 01014 (88 words).

10102 10132 13132 13232 10232 30232 30132 32132 12132 12130
32130 30130 10130 10230 30230 20230 23230 13230 13130 23130
20130 20131 23131 23231 20231 30231 30131 32131 32101 32301
30301 20301 20101 30101 30201 20201 23201 23101 23121 20121
30121 32121 32321 30321 20321 20323 10323 12323 12320 32320
30320 10320 20320 20120 23120 13120 12120 32120 30120 10120
10123 12123 13123 23123 20123 20103 20203 20303 10303 10302
30302 32302 12302 12303 12103 10103 10203 13203 23203 23103
13103 13102 13202 10202 30202 30102 32102 12102

Cycle for 01024 (85 words).

10102 10202 30202 30212 10212 13212 13210 23210 20210 30210
10210 10213 10203 20203 23203 13203 13213 23213 20213 20313
10313 12313 12312 12310 32310 30310 20310 10310 10312 30312

32312 32302 30302 10302 10303 20303 20103 23103 13103 10103
12103 12303 12302 12102 32102 30102 30101 30201 20201 23201
23101 20101 20301 30301 32301 32101 32131 30131 30231 20231
23231 23131 20131 20130 23130 13130 13230 23230 20230 30230
10230 10130 30130 32130 12130 12132 32132 30132 30232 10232
10132 13132 13232 13202 13102

Cycle for 01104 (74 words).

10210 10310 20310 30310 32310 12310 12312 10312 30312 32312
32012 32032 32031 32021 32321 30321 20321 20320 10320 30320
32320 12320 12020 12010 32010 32020 32030 12030 12032 13032
13232 13212 13012 12012 12013 12313 10313 20313 20323 10323
12323 12023 13023 23023 23013 13013 13213 23213 20213 10213
10212 10232 10230 13230 23230 23030 13030 13020 23020 23021
23031 23231 20231 30231 30232 30212 30210 30230 20230 20210
23210 23010 13010 13210

Cycle for 01124 (74 words).

10202 10203 20203 23203 23201 20201 20301 20303 10303 12303
12302 32302 32301 30301 30201 30231 20231 20230 30230 10230
13230 23230 23231 23031 23030 13030 12030 32030 32031 32032
32012 32010 12010 12012 12032 13032 13232 10232 30232 30202
30212 10212 10213 10210 30210 20210 20310 30310 10310 12310
32310 32312 12312 12313 12013 13013 23013 23010 23210 13210
13010 13012 13212 13202 13203 13213 23213 20213 20313 10313
10312 30312 30302 10302

Cycle for 01204 (83 words).

10120 20120 23120 13120 13130 23130 20130 10130 30130 32130
12130 12132 32132 30132 10132 13132 13032 12032 32032 32012
12012 13012 13010 12010 32010 32020 12020 13020 23020 23010
23030 13030 12030 32030 32031 32131 30131 20131 23131 23031
23021 23121 20121 30121 32121 32021 32321 30321 20321 20320
10320 30320 32320 12320 12323 12023 13023 23023 23013 13013
12013 12313 12310 32310 32312 12312 10312 30312 30310 10310
20310 20313 10313 10323 20323 20123 23123 13123 10123 12123
12120 32120 30120

Cycle for 01234 (82 words).

10102 10302 30302 32302 12302 12303 10303 10323 20323 20303
20103 10103 12103 12123 13123 23123 20123 10123 10120 30120
32120 12120 13120 23120 20120 20320 10320 30320 32320 12320
12310 32310 30310 10310 20310 20313 10313 10312 30312 32312
12312 12313 12323 12023 13023 23023 23013 23010 13010 12010
12013 13013 13012 12012 32012 32010 32020 12020 13020 23020
23021 23121 20121 30121 32121 32021 32321 30321 20321 20301
30301 32301 32101 30101 20101 23101 23103 13103 13102 12102
32102 30102

Pool for 00000:0.

12123-13123 12312-12313 21231-31231 23121-23131 31212-31312

Pool for 00000:1.

12132-13132 13212-13213 21213-21313 21321-31321 32121-32131

Pool for 00001:0.
21210-21213 21230-31230

Pool for 00011:0.
12102-13102 13102-13103 13123-23123 13130-13230
13202-13203 21203-21303 31202-31302 32130-32132

Pool for 00012:0.
13130-13230 21301-21321 23120-23130 23121-23123
31201-31301 32101-32301 32121-32131 32121-32321

Pool for 00101:0.
12010-12020 12012-13012 12013-12023 12320-12323
13013-23013 13210-13212 21010-31010 23010-23210

Pool for 00102:0.
12013-13013 12313-12323 13013-23013 21013-21213
21023-21323 21031-23031 23020-23021 31031-31231

Pool for 00112:0.
21020-23020 21030-31030

Pool for 00121:0.
12010-13010 21010-21030 21010-31010 21210-21213
23010-23210 23030-23230 31012-31312 31030-32030

Pool for 00123:0.
13030-23030 13230-23230 21010-21310 21030-31030
21230-21231 31010-31210 31012-32012 31030-31230

Pool for 01012:0.
10103-13103 10123-13123 10130-30130 13130-13230
20101-30101 20230-30230 20301-20321 30131-30231

Pool for 01023:0.
12130-13130 20101-30101 20201-20301 20210-30210
20231-30231 23130-23131 30102-32102 30202-30212

Pool for 00004:0.
12312-12313 13130-13132 13212-13213 21210-21213
21321-21323 23121-23123 31231-31232 31310-31312

Pool for 00014:0.
12102-32102 12103-13103 12120-32120 12130-12132
13130-23130 21201-21203 23121-23123 32101-32121

Pool for 00104:0.
13012-13013 13210-23210 13212-13213 21020-21023
21320-21321 21321-21323 31320-31321 32320-32321

Pool for 00114:0.
12030-13030 21020-21021 21301-21321 21320-21323
23021-23023 31020-31021 31020-32020 32021-32031

Pool for 00124:0.
12310-32310 13210-13212 21030-31030 21201-31201
21213-21313 23010-23210 31010-32010 31301-31302

Pool for 01014:0.
10120-10123 10202-13202 10302-10303 12120-13120
12123-13123 13102-13103 20130-20131 20321-20323

Pool for 01024:0.
10103-13103 10132-13132 10302-10303 10312-30312
12130-12132 12312-12313 20103-20303 20213-20313

Pool for 01104:0.
10210-10310 12010-12020 12012-12013 12310-12312
13013-13213 20210-23210 23030-23230 32012-32312

Pool for 01124:0.
10202-10203 10210-10213 10230-13230 10310-30310
13010-13210 20210-20310 23230-23231 32310-32312

Pool for 01204:0.
10123-13123 12010-32010 12120-12123 12310-12313
12320-12323 23010-23020 30310-30312 32030-32031

Pool for 01234:0.
10102-30102 12102-32102 12123-13123 12302-32302
12310-12320 13012-13013 13102-13103 20101-30101

The thirty constant-source interfaces. Write $K_i = E(C_i) \setminus O_i$ for the two constant components. Coordinate rotation fixes both components, since its order is five. Each five-edge pool O_i is a rotation orbit, so K_i is invariant under $\rho_r(x)_k = x_{k+r \pmod 5}$. The following edges belong to the unchanged protected set K_H of type 00001:

$$a = 31212 - 31232, \quad b = 12132 - 32132, \quad c = 21313 - 21323, \quad d = 13123 - 23123.$$

When boundary entry j changes from 0 to $q \in \{1, 2, 3\}$, its canonical normalizer is one of

$$N_{j,\epsilon}(x)_k = p_q(x_{j-4\epsilon+\epsilon k}), \qquad \epsilon \in \{+1, -1\},$$

where p_q is the first lexicographic permutation with $p_q(0) = 0$ and $p_q(q) = 1$. Indeed, the four repeated entries map to 0 and the unique changed entry maps to 1 at position 4.

The table gives $e_{i,q} \in K_i$ and its two normalized images; these memberships follow directly from the printed words and pools.

i	q	$e_{i,q}$	$N_{0,+1}(e_{i,q})$	$N_{0,-1}(e_{i,q})$
0	1	$23121-23123$	a	b
0	2	$31231-31232$	c	d
0	3	$21231-23231$	d	c
1	1	$21213-23213$	b	a
1	2	$31321-32321$	d	c
1	3	$21321-21323$	c	d

At any position j, use $\rho_{-j}(e_{i,q}) \in K_i$. Its image is

$$N_{j,\epsilon}(\rho_{-j}(e_{i,q})) = N_{0,\epsilon}(e_{i,q}) \in K_H,$$

so it lies in both actual protected sets. These six families cover all thirty interfaces.

Fixed-side source selectors (236 interfaces). Let K be either fixed protected set and B the opposite component, with pool P_B and a Hamilton cycle. At a common state x, put $d = \deg_B(x)$ and $g = |(K \cap \delta_B(x)) \setminus P_B|$. If $d - g \leq 1$, every Hamilton cycle in B takes two incident edges at x, at least one in $K \setminus P_B$. Only B needs a Hamilton cycle.

t	S_t
00001	13123, 21230, 23120, 32132
00004	12130, 13123, 32312
00011	12323, 21230, 21320
00012	12303, 13203, 13230, 31230
00014	12303, 31202, 31230
00101	12312, 13230, 13232, 21323, 31312
00102	13023, 13210, 13230, 23231, 31231, 32031, 32321
00104	12032, 12323, 13232, 23021, 23031
00112	12023, 13230, 31201
00114	12032, 21303
00121	12302, 12303, 21203, 21303
00123	12302, 12310, 23201, 32031
00124	12303, 23031, 32301
01012	10203, 13203, 23231, 32301
01014	12302, 23201, 23231, 32301
01023	12310, 13202, 20310, 23101
01024	23213, 32301, 32310
01104	20231, 30321, 32021
01124	12303, 23031, 32301
01204	10132, 13032, 23013, 30312, 32312
01234	23013, 23101

For each of the 236 nonconstant-source interfaces not listed individually below, some $x \in S_t$ lies in both actual components and satisfies $d - g \leq 1$, with fixed source K against the restored target component, or restored target K against the source component. The opposite side has its displayed

Hamilton cycle; the printed protected sets and pools use the original interface's first normalizer. Its degree is

$$L_i(b,x) = \{0,1,2,3\} \setminus \{x_{i-1}, x_{i+1}, b_i\}, \qquad d = \sum_i (|L_i(b,x)| - 1),$$

where b is its boundary in the original coordinates, omitting b_i when starred. This proves the specified-template identities in all 236 cases.

The 47 remaining individual nonconstant-source interfaces.

```
00001:0 1 2 00102:0 21313-21323 ; 00001:0 1 3 00102:0 21313-21323
00001:0 2 1 00101:0 12310-32310 ; 00001:0 4 2 00001:0 13120-23120
00001:0 4 3 00001:0 12310-32310 ; 00011:0 0 2 00112:0 12130-12132
00011:0 0 3 00112:0 12120-12130 ; 00101:0 0 1 00101:0 21020-23020
00101:0 1 1 00101:0 12013-12313 ; 00101:0 2 0 00001:0 12310-32310
00101:0 3 2 00121:0 12010-32010 ; 00101:0 3 3 00121:0 12010-32010
00101:0 4 0 00001:0 12032-32032 ; 00102:0 0 1 01012:0 21013-23013
00112:0 0 1 00102:0 21030-21230 ; 00112:0 0 2 00112:0 12023-13023
00112:0 1 3 00123:0 12320-32320 ; 00112:0 2 2 00121:0 12023-13023
00112:0 2 3 00123:0 13020-13023 ; 00112:0 3 2 00112:0 13030-13230
00112:0 4 3 00112:0 12320-32320 ; 00121:0 2 2 00112:0 12010-32010
00121:0 3 0 00101:0 12010-32010 ; 00121:0 4 2 00112:0 12013-13013
00123:0 2 2 00112:0 12030-32030 ; 01012:0 0 1 00102:0 20103-23103
01012:0 1 2 01012:0 10120-13120 ; 00104:0 0 1 01014:0 21010-31010
00104:0 1 2 01204:0 13010-23010 ; 00104:0 1 3 01204:0 12010-32010
00124:0 2 3 00124:0 12010-32010 ; 01014:0 0 1 00104:0 20101-30101
01014:0 0 2 01024:0 10102-12102 ; 01014:0 0 3 01024:0 10103-10203
01014:0 1 2 01024:0 10120-30120 ; 01014:0 1 3 01024:0 10102-12102
01014:0 2 2 01024:0 10102-10132 ; 01014:0 2 3 01024:0 10102-10132
01024:0 0 3 01234:0 10103-12103 ; 01024:0 1 2 01014:0 10102-13102
01024:0 2 3 01234:0 10102-13102 ; 01204:0 1 0 00104:0 12012-13012
01204:0 1 3 01204:0 10120-30120 ; 01234:0 0 2 01024:0 10102-10302
01234:0 1 3 01024:0 10102-10302 ; 01234:0 2 0 01024:0 10103-12103
01234:0 3 1 01024:0 10102-10302
```

E Shared-hub path certificates

Let $v_0, \ldots, v_{r-1}$ be the restored path, where $r \in \{4, 5\}$. Its endpoints have the same outside neighbour w; the other outside neighbours of its internal vertices are distinct and different from w. In the physical port order $t = (a, b_1, \ldots, b_{r-2})$, the fibre consists of the proper path words with coordinate exclusions

$$(a, b_1, \ldots, b_{r-2}, a).$$

All colours in this appendix and its arrays lie in $\{0, 1, 2, 3\}$. The files `data/path4-family.json` and `data/path5-family.json` supply literal cyclic arrays, one for each normalized boundary. Normalize colours in first-appearance order, then append unused colours in ascending order. Restore every cycle by the inverse palette map, without reflecting the path or choosing an additional relative map.

Table E.1: Complete shared-hub path coverage. Interfaces are directed.

Order	Templates	States	Boundaries	Interfaces	Min. edges
4	5	160	64	576	3
5	15	1112	256	3072	5

Here "States" sums the lengths of the normalized arrays. The verifier reconstructs each legal fibre independently, then checks complete coverage, no repeated state, and a one-coordinate change at every cycle step, including the closing step.

The verifier expands to every labelled boundary and intersects the *chosen cycle edge sets* at every physical one-port change. A change at w changes both endpoint exclusions simultaneously; it is one physical interface, included among the 576 or 3072 directed cases. The endpoints of every shared edge are legal on both sides, so both matching outside-recolouring edges exist. The displayed minima exceed the two edges needed to select an outgoing splice after one incoming edge has been removed.

The verifier `verify_shared_hub_paths.py` checks these finite claims, including complete template coverage, unique states, legal colour words, and the stated interface minima. Splicing along an arbitrary bottom Hamilton path and the article's structural induction remain analytic arguments; neither is inferred from finite replay.

F Degree-two configuration families

The seven files `data/degree-two/`*kind*`-family.json` give literal cycles for the following fixed internal models. Paths have vertices $0, \dots, n-1$ in order; the star has centre 0. Write T_m for the cycle $0, \dots, m-1, 0$ with the additional vertex m and edge $0m$. A boundary is $t \in \{0,1,2,3\}^q$. At vertex v, exclude the colours t_i for $i \in I_v$. In the tuples below a single index denotes its singleton set. An empty set excludes no colour; a two-element set can exclude one or two colours, according to their physical equality.

Kind	X	$(I_0, \dots, I_{n-1})$
`star3`	$K_{1,3}$	$(0,1,2,3)$
`path4heavy`	P_4	$(0,\{1,2\},3,0)$
`path5heavynear`	P_5	$(0,\{1,2\},3,4,0)$
`path5heavycenter`	P_5	$(0,1,\{2,3\},4,0)$
`path6`	P_6	$(0,1,2,3,4,0)$
`tailed5`	T_5	$(0,\varnothing,1,2,3,4)$
`tailed6`	T_6	$(0,\varnothing,1,2,3,4,5)$

The port counts q are respectively $4,4,5,5,5,5,6$. In every path model, port 0 is the common outside neighbour of both endpoints, so its move changes both endpoint exclusions at once.

Two port roles can represent one outside vertex only if every pair of their incident internal vertices has distance at least three; otherwise a triangle or a 4-cycle results. The complete additional alias pairs are

`path6` : $(1,4)$,
`tailed5` : $(1,4)$, $(2,4)$,
`tailed6` : $(0,2)$, $(1,4)$, $(1,5)$, $(2,5)$, $(3,5)$.

The first four models have none. Each compatibility graph is triangle-free, so no outside vertex occupies three roles. Several disjoint pairs may be identified. A move changes one singleton role or one listed pair, with equal old entries in the latter case.

Table F.1: Degree-two family coverage; interfaces are directed.

Kind	Templates	States	Boundaries	Interfaces	Min. edges
star3	15	499	256	3072	4
path4heavy	15	376	256	3072	2
path5heavynear	51	2915	1024	15360	2
path5heavycenter	51	2921	1024	15360	2
path6	51	8593	1024	16128	9
tailed5	51	8535	1024	16896	12
tailed6	187	72197	4096	89088	19

The primary verifier reconstructs every legal fibre from the fixed models, checks the literal cycles and verifies alias completeness. Every labelled singleton or paired transition is enumerated. Palette normalization is as in Appendix E. An independent verifier restores cycles to physical labels and checks the corresponding selected-edge sets.

Both verifiers reconstruct the seven fixed models, verify their legal states, cycles, and alias transitions, and reproduce Table F.1. Their checks cover 421 templates, 96,036 template states, 8,704 boundaries and 158,976 directed interfaces. The margin of two is the finite premise for restoration; the unbounded splice and structural inductions remain analytic proofs.

G Pentagonal reservoirs and full-list families

The three new arrays in `data/pentagonal-reservoirs/` have the following fixed models, using the exclusion notation of Appendix F. The cycle is $0, 1, 2, 3, 4, 0$; vertex 5, when present, is the additional degree-two vertex. The model R deletes an ordinary pentagon and a nonadjacent degree-two vertex; F deletes a pentagon with one degree-two vertex; P deletes an ordinary pentagon and an adjacent degree-two vertex.

Model	X	$(I_0, \ldots, I_{n-1})$	q
R	$C_5 \dot\cup \{5\}$	$(0, 1, 2, 3, 4, \{5, 6\})$	7
F	C_5	$(\varnothing, 0, 1, 2, 3)$	4
P	$C_5 + 05$	$(\varnothing, 0, 1, 2, 3, 4)$	5

The file for R is `reservoir-family.json`; the files for F and P are `pentagon-full-family.json` and `pentagon-pendant-family.json`, respectively. Their palette is $\{0, 1, 2, 3\}$, with first-appearance normalization as before. The two neighbours of vertex 5 in model R are distinct, but their colours may agree.

The complete alias pairs for R are (i, j) with $0 \le i < 5$ and $j \in \{5, 6\}$. They form $K_{5,2}$, with $1 + 10 + 20 = 31$ matchings. Model F has no aliases; model P has pairs $(1, 4)$ and $(2, 4)$, hence three matchings. The internal-distance test and absence of compatible triples are checked independently by both verifiers. Several disjoint identifications introduce no additional move type: an actual recolouring changes one role or one identified pair whose old entries are equal.

Table G.1: Pentagonal family coverage. Moves and component overlaps are directed.

Kind	Types	Cycles	States	Labels	Moves	Overlaps	Min. edges
reservoir	715	720	91410	16384	466944	470208	4
full	15	15	1105	256	3072	3072	11
pendant	51	51	8606	1024	16896	16896	8

In model R, a fibre has two components exactly when its first five boundary entries agree; otherwise it is connected. There are five such normalized disconnected boundaries. Each displayed cycle covers one *whole component*, not the disconnected fibre. The smallest component has order 30. Models F and P have connected fibres. The primary verifier enumerates all legal states and their components; an independent verifier reconstructs these states and components separately. Both check exact component coverage, cycle lengths, uniqueness and all closing steps.

At every labelled interface, *every overlapping component pair* has the stated common-cycle-edge margin. Every component on either side overlaps a component on the other side, including at shared-port moves. The independent verifier transports literal cycles into physical labels and checks the corresponding selected-edge sets. The verifiers agree on the full common-edge histograms. In total the new data contain 781 boundary types, 786 component cycles and 101,121 states, covering 17,664 labels and 486,912 directed moves.

Margin four supplies the reservoir tree-splicing reserve: along a bottom path with at most two components per layer, a component-graph node has degree at most four and loses at most three edges before its next attachment. For the connected families, margin two suffices along a bottom path. Unchanged boundaries retain the identical selected cycles, whose lengths are also checked against these capacities. The connected-layer argument, unbounded splicing and structural inductions are analytic proofs in the article. The redundant alternating-hexagon family and exploratory full-graph tests are not premises of these certificates.

H Linked pentagons and a three-contact path

The five compressed arrays in `data/extended-cycles/` are literal Hamilton cycles for fixed local graphs. Their palette is $\{0, 1, 2, 3\}$. A role at a vertex excludes its boundary colour; vertices with no role have full lists. The two shared-edge models have cycles $(0, 1, 4, 3, 2, 0)$ and $(0, 1, 7, 6, 5, 0)$, with the optional edge 36. Their role orders are respectively $(2, 3, 4, 5, 6, 7)$ and $(2, 4, 5, 7)$. The three-contact path has vertices $0, \ldots, 6$ and exclusion pattern (a, b, c, a, d, e, a), so its retained hub is one role with three contacts.

The linked models have cycles 012340 and 567895, with edge 05 and, in the two-link case, edge 27. Their roles are the vertices without a joining edge, in increasing order. Thus they have eight and six roles. The literal files have the kind names in Table H.1, followed by `.json.gz`.

The shared-edge and path models use first-appearance palette normalization, appending unused colours increasingly. The one-link and two-link models also minimize over their eight and four declared graph automorphisms. Each automorphism is checked to preserve edges and contact sets; inverse spatial and palette maps restore physical coordinates. Both verifiers use the fixed graph and contact models, cover all canonical boundaries, and verify exact cyclic coverage.

Two roles may coincide precisely when all pairs of their contacts have internal distance at least three. The complete extra pairs are

shared edge	$(0,4), (0,5), (1,3), (1,4), (1,5), (2,3), (2,4),$
shared edge with chord	$(0,3), (1,2),$
three-contact path	$(1,3), (1,4), (2,4).$

For the linked models they are all pairs between the two cycles' role sets. The compatibility graphs have no triangle, hence no three distinct roles can coincide. Their numbers of complete alias matchings are respectively $22, 4, 5, 209, 34$. Several disjoint aliases are allowed, while each actual move changes one role or one equal pair simultaneously. The repeated hub role changes three local exclusions at once.

Table H.1: Extended cycle families; interfaces are directed physical moves.

Kind	Types	States	Labels	Interfaces	Min. edges
sharededge	187	158054	4096	95232	56
sharededgechord	15	16958	256	3456	94
threecontactpath7	51	18843	1024	17664	10
onelinkpentagons	451	1917512	65536	2359296	219
twolinkpentagons	67	379795	4096	101376	390

In total there are 771 templates with 2,491,162 states, covering 75,008 labelled boundaries and 2,577,024 directed interfaces. The primary verifier checks legal states and all selected-cycle intersections. An independent verifier reconstructs the fixed models and maps, restores physical colour strings, and reproduces every interface histogram. Both check the displayed counts and all selected-cycle intersections. Only a margin of two is used in the analytic path-order splice.

I Protected bowtie seeds and clique trees

A bowtie has cycles 012340 and 056780, with vertex 0 of original degree four and the other vertices of original degree three. The chorded model adds edge 26. Their exterior role orders are $(1,2,3,4,5,6,7,8)$ and $(1,3,4,5,7,8)$; all unlisted vertices have full lists. In `data/bowties/`, the two seed arrays have suffix `-seeds.json.gz`, and the two tree arrays have suffix `-clique-trees.json.gz`; their prefixes are `bowseed` and `bowchord`. The file `ordinary-pentagon-seeds.json` extracts the ordinary seeds from Appendices C–D.

Palette normalization is by first appearance, with unused colours appended increasingly, after minimizing over the eight bare or two chorded spatial maps. The fixed maps preserve the graph and exterior roles. Canonical boundaries and inverse maps are reconstructed by the verifiers before the seed checks.

For a bowtie fibre B_t, the seeds are cycles of $B_t \square K_2$ and $B_t \square K_3$. They use the complete clique row adjacency, including moves between rows zero and two. Every edge of a listed matching M_t is retained in every physical row. The first two matching edges are distinct reserves; the rest form the protected core. Every cycle is checked for complete coverage, uniqueness, legality and closure. An independent verifier reconstructs the same finite graphs and restores physical coordinates directly.

Table I.1: Bowtie seeds and interfaces. The last columns give full-matching and core overlap minima after deleting the two reserves.

Kind	Types	Fibre states	Seed states	Interfaces	Full	Core
bowseed	451	640608	3203040	2162688	9	9
bowchord	107	204832	1024160	89088	25	24

There are 1,116 clique-product seed cycles, with 4,227,200 seed states. The complete possible shared-role pairs number twelve and five; neither compatibility graph has a triangle. Both verifiers check all 65,536 and 4,096 labelled boundaries, and all 2,162,688 and 89,088 directed factor interfaces. They reproduce full-matching and core histograms. These factor checks also handle outside vertices shared between distinct deleted factors; their combined interfaces factorize analytically.

The clique arrays partition every bowtie fibre into edges and triangles. Every block edge is a legal one-vertex move. The listed connectors form a tree on the blocks, and at most two meet each physical fibre vertex. They cover 558 types and 845,440 states, with 422,601 blocks including 238 triangles, and 422,043 connectors. A three-vertex path does not meet the triangle contract.

The ordinary extract has eleven boundary types, twelve components and 590 component states, with 24 two- and three-row seeds. The verifier checks its exact cycles, pools, protected sets, reserves and hard seeds against the marked blocks in Appendices C–D. This preserves the relative-map protection contract used in the article. Its P_3 seeds also give K_3 seeds because $P_3 \subseteq K_3$; their protected sets need not be matchings.

The generalized protected-product lemma, its use over arbitrary bottom paths, and the completed-dual structural proof remain analytic.

J Hinged-pair product seeds and clique trees

A hinged pair has cycles $(0,2,3,4,1,0)$ and $(0,1,7,6,5,0)$. Vertex 0 has original degree four and every other vertex has original degree three. The chorded model adds edge 36. The bare exterior-role order is $(0,2,3,4,5,6,7)$, and the chorded order is $(0,2,4,5,7)$. Each role excludes one outside colour; all unlisted vertices have full lists. In particular the heavy vertex has one outside exclusion.

The compressed files in `data/hinged-pairs/` are `cycles.json.gz` and `trees.json.gz`; they describe the fixed models `bare` and `chord`. Boundaries are seven- or five-digit strings over $\{0,1,2,3\}$. Matching words have eight digits; seed words prepend the physical row. Tree indices are zero-based in ascending lexicographic order of legal eight-digit words.

Normalize over the identity and arm exchange $(0,1,5,6,7,2,3,4)$, renaming colours by first appearance and appending unused colours increasingly. The least full normalization tuple fixes ties. Each map preserves the graph and role contacts; its inverse restores physical coordinates. The complete compatible role pairs are

$$\begin{aligned} \text{bare} &\quad (1,5),(1,6),(2,4),(2,5),(2,6),(3,4),(3,5),\\ \text{chorded} &\quad (1,4),(2,3). \end{aligned}$$

They are exactly the pairs at internal distance at least three. There is no compatible triple. The complete alias matchings number 22 and 4; a move changes one role or one equal pair simultaneously.

Every template supplies a matching M and Hamilton cycles in its fibre product with K_2, K_3 and P_3. Each matching edge is retained in every physical row of all three seeds. The first two edges are reserves, and the remaining edges form the protected core. A K_3 seed may use row edges 02; a P_3 seed may only change between adjacent rows.

Table J.1: Hinged-pair seeds and directed factor interfaces. Full and core columns give the respective overlap minima.

Kind	Types	Fibre states	Seed states	Interfaces	Full	Core
bare	379	241774	1934192	430080	6	6
chord	31	26051	208408	16896	21	20

The 410 templates have 267,825 fibre states and 1,230 separate seed cycles with 2,142,600 seed states. Both verifiers reconstruct complete legal state sets, check every seed's coverage, legality, closure and row retention, and verify all 17,408 labelled boundaries and 446,976 directed interfaces. An independent verifier reconstructs the graph, contact, spatial and alias contracts, restores literal physical strings at every boundary, and reproduces the complete full/core histograms. The fixed bare boundary $(2,2,1,1,3,1,1)$ is also reconstructed: it has 526 states, 1,578 edges and three leaves, excluding a Hamilton path in that single fibre.

The primary verifier checks all 410 clique trees, with 133,811 blocks, including 203 triangles, and 133,401 connectors. Every block is an edge or a genuine triangle; connectors give a quotient tree of physical capacity two.

The article uses the K_2/K_3 seeds and clique trees. Ordinary-pentagon partners use the existing ordinary seeds first and the hinged clique tree second, preserving the relative-map contract of Appendices C–D. The P_3 seeds are additional finite witnesses. Products, combined interfaces, unbounded restoration and structural counts remain analytic.